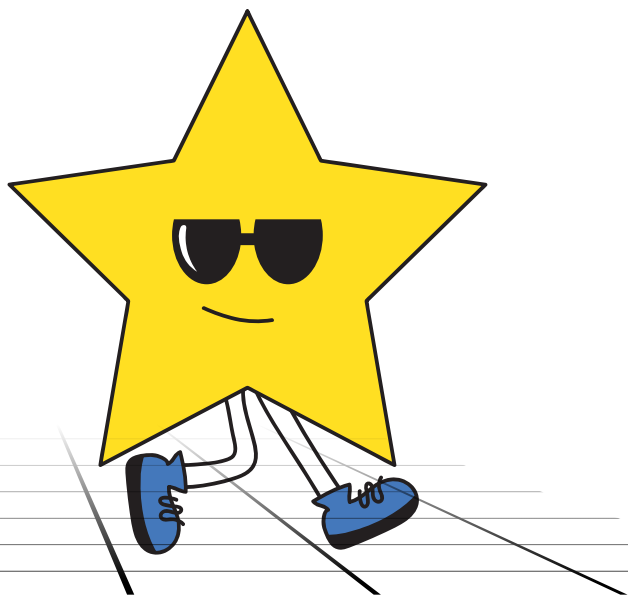
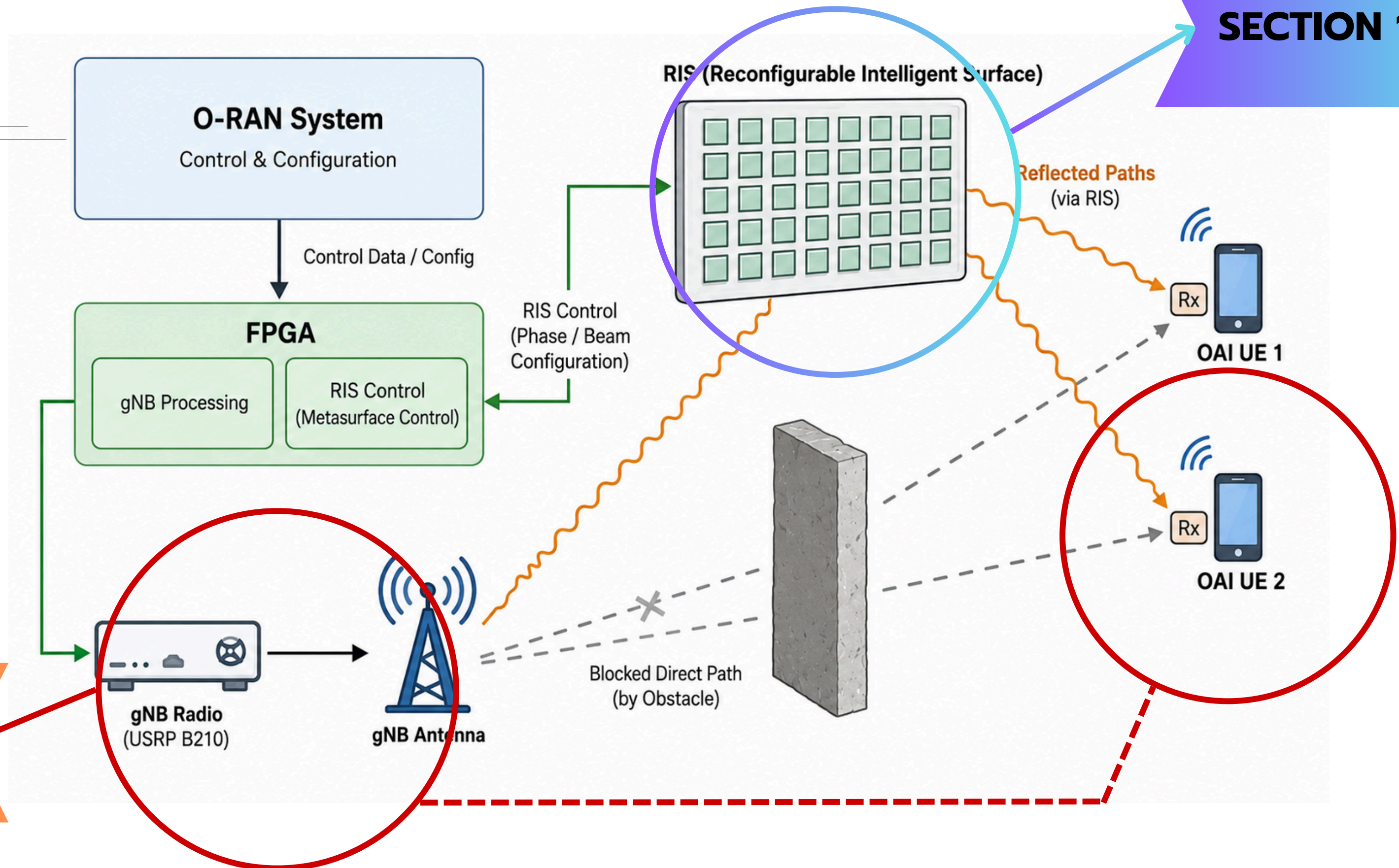




# OVERVIEW

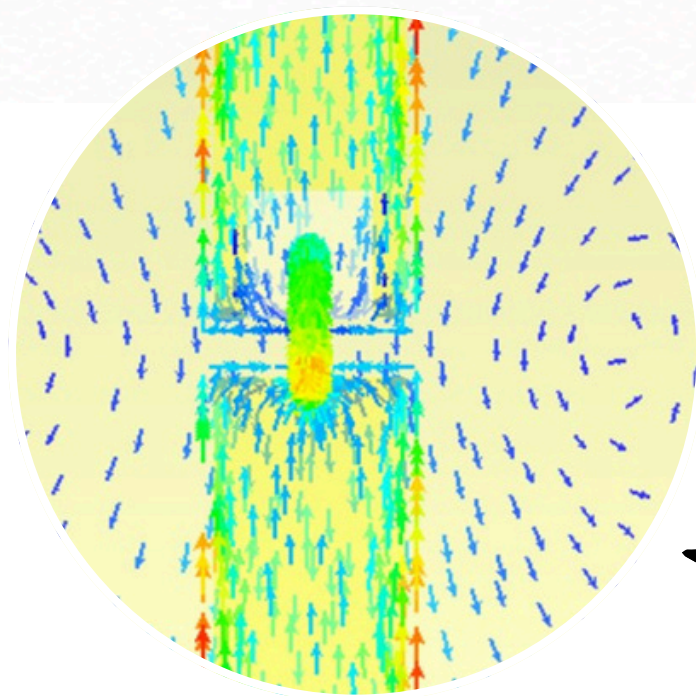
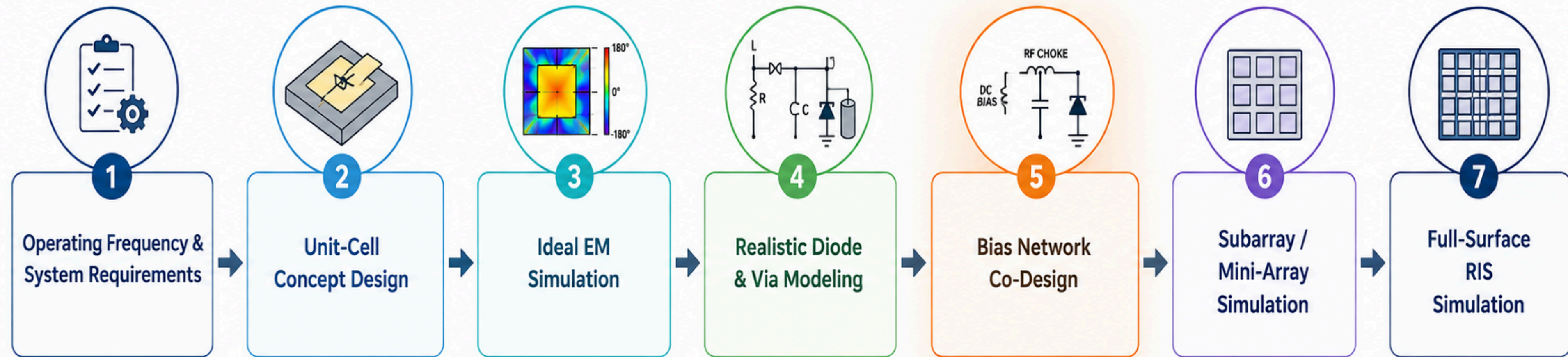
## SECTION 1



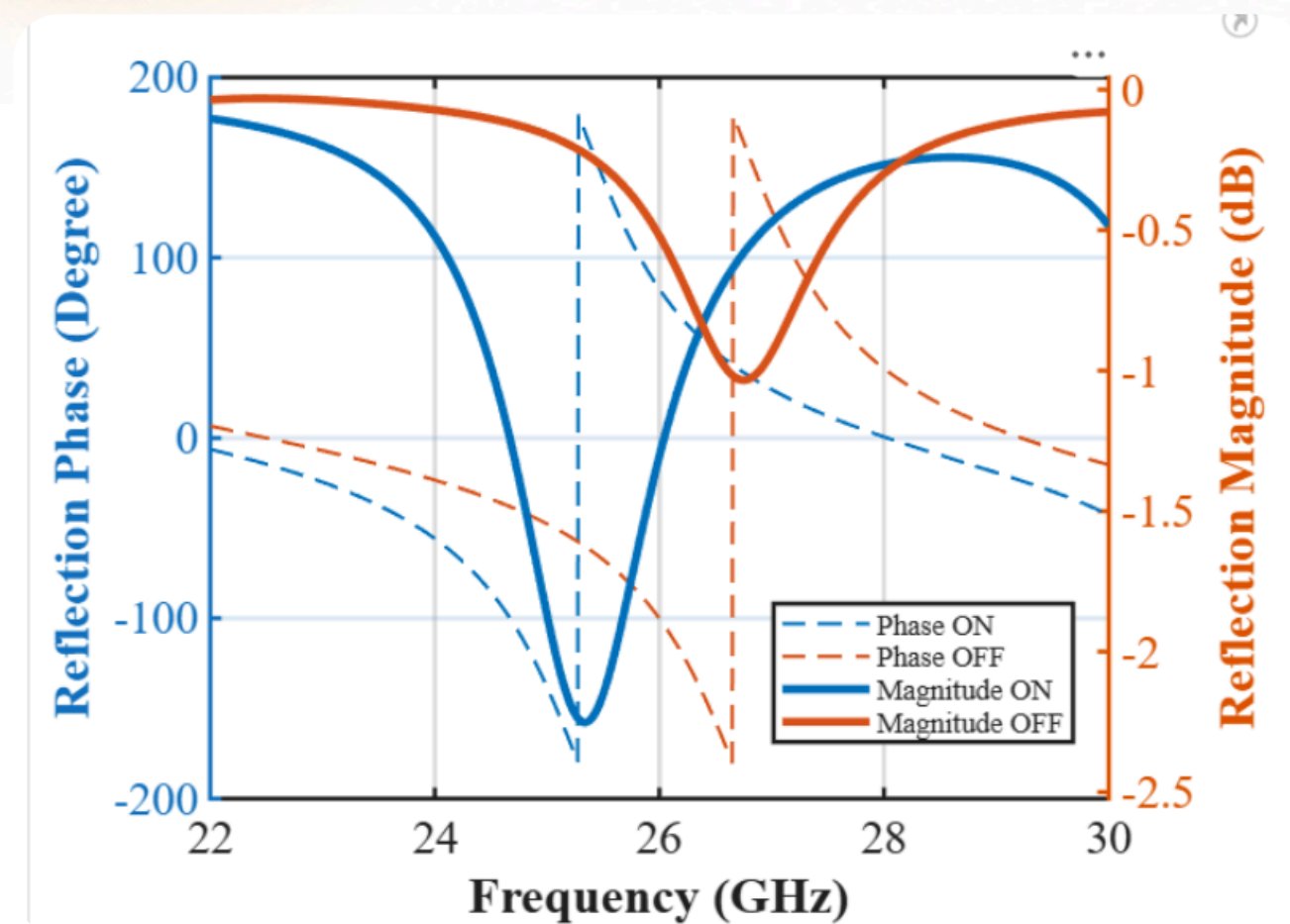
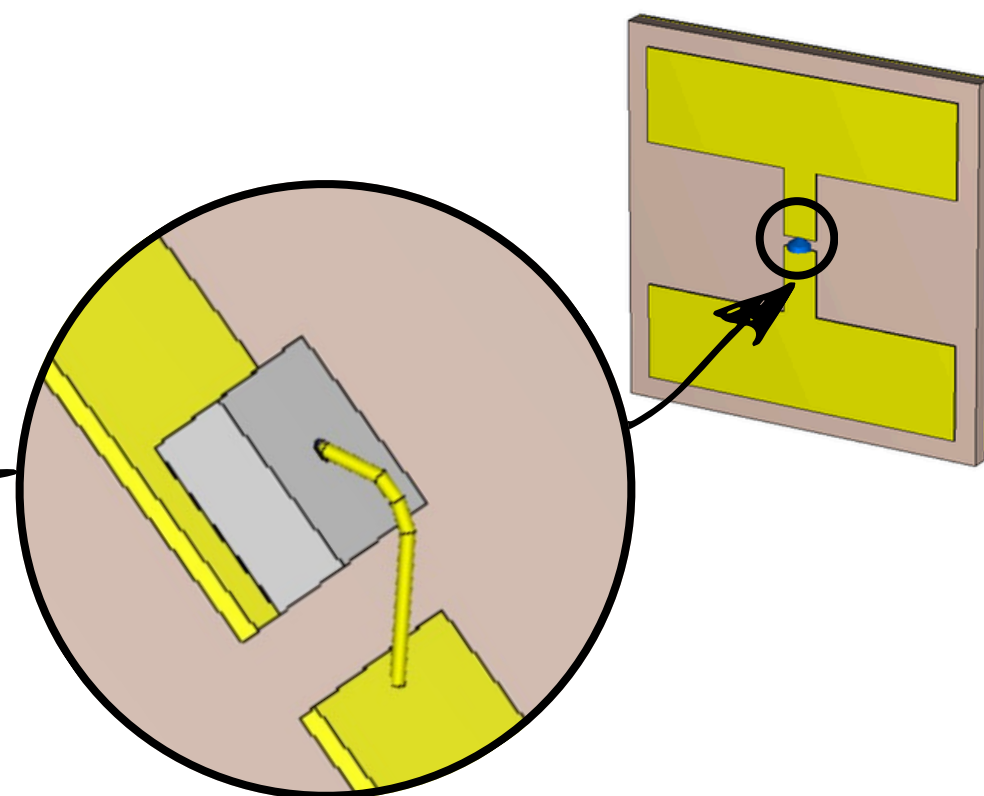
## SECTION 2

## SECTION 1

## 1-BIT RIS DEVELOPMENT WORKFLOW



**Surface Current**

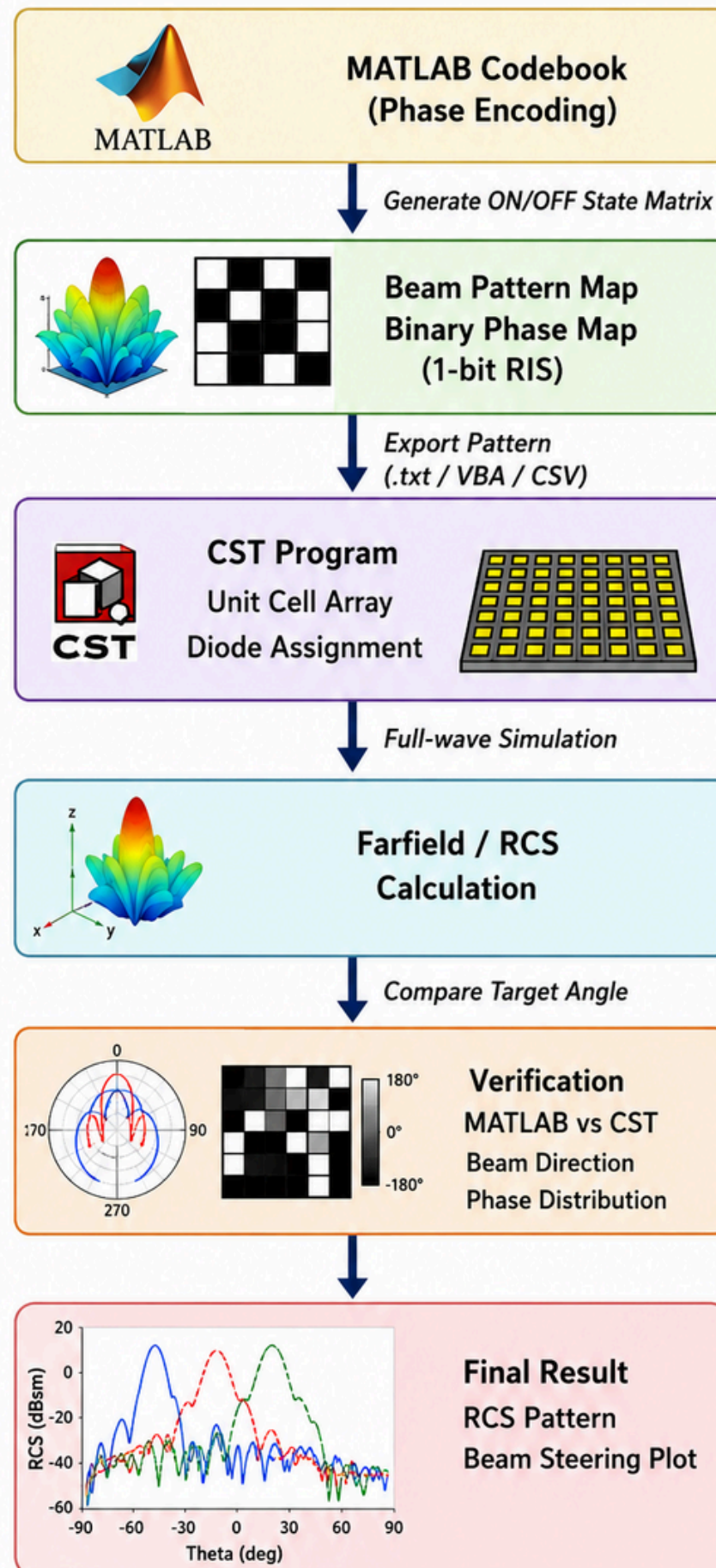


# Full Array

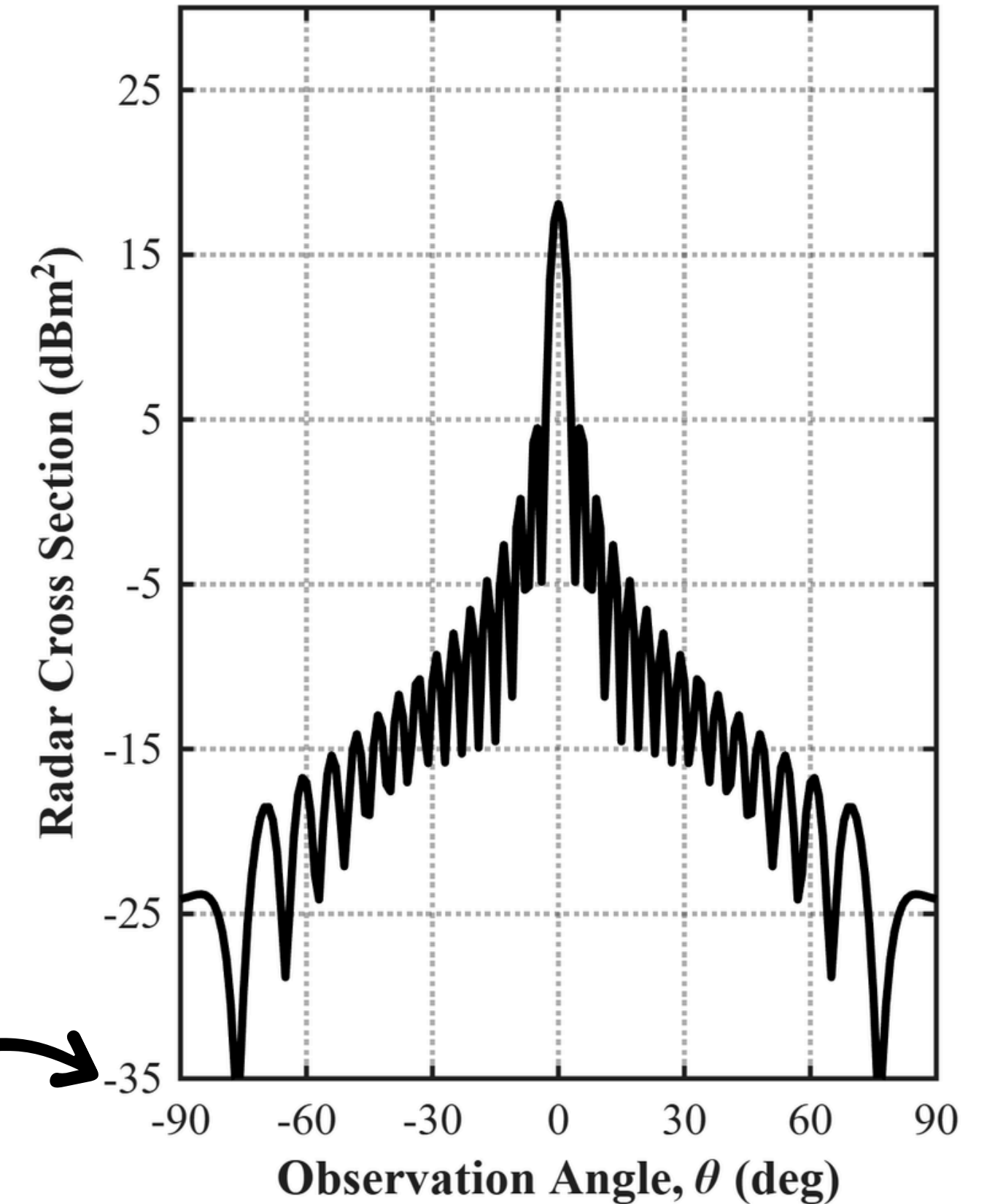
[In Progress...]

Codebook  
Verification

## RIS / Reflectarray Workflow

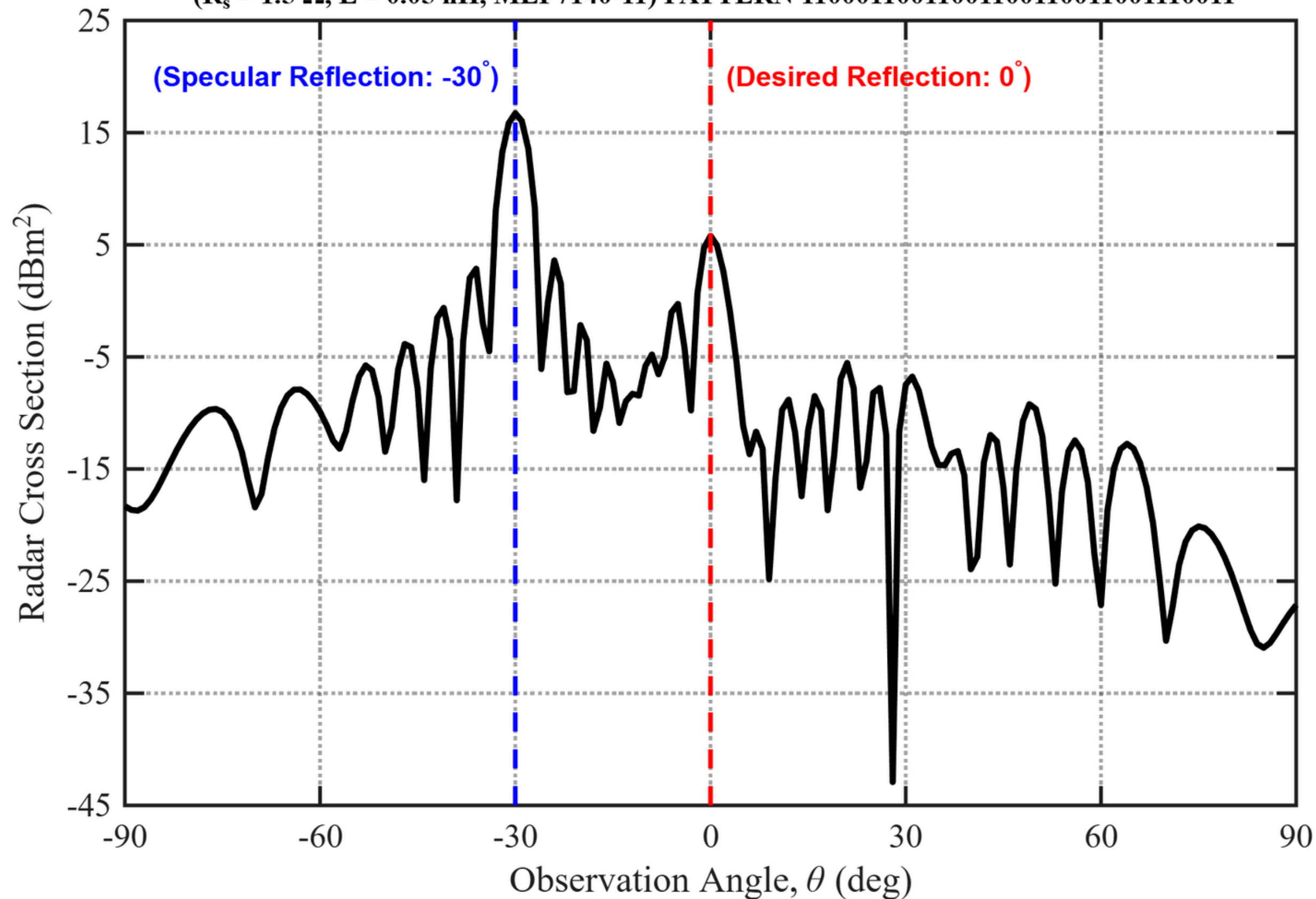


Radar Cross Section of  $1.76 \times 1.76$  cm RIS  
on RT 6010 @ 26 GHz (  $0^\circ$  Incidence )  
All ON Mode ( $R_s = 1.5 \Omega$ ,  $L = 0.05$  nH, MLP7140-11)



# CST Simulation Results for 32\*32 Array

Radar Cross Section of  $1.76 \times 1.76$  cm RIS for RT 6010 @ 26 GHz (  $30^\circ$  Incidence )  
( $R_s = 1.5 \, \Omega$ ,  $L = 0.05$  nH, MLP7140-11) PATTERN 11000110011001100110011001110011



## Simulation & Wave Config.

### [Excitation & Wave Property]

Excitation Type: Plane Wave

Polarization: Linear Polarization

E-Field Vector: ( $x=0$ ,  $y=1$ ,  $z=0$ )

Wave Direction: ( $-0.5$ ,  $0$ ,  $-0.866$ )

Incident Angle:  $30^\circ$  Oblique

### [Numerical Solver & Mesh]

Solver: Time Domain Solver

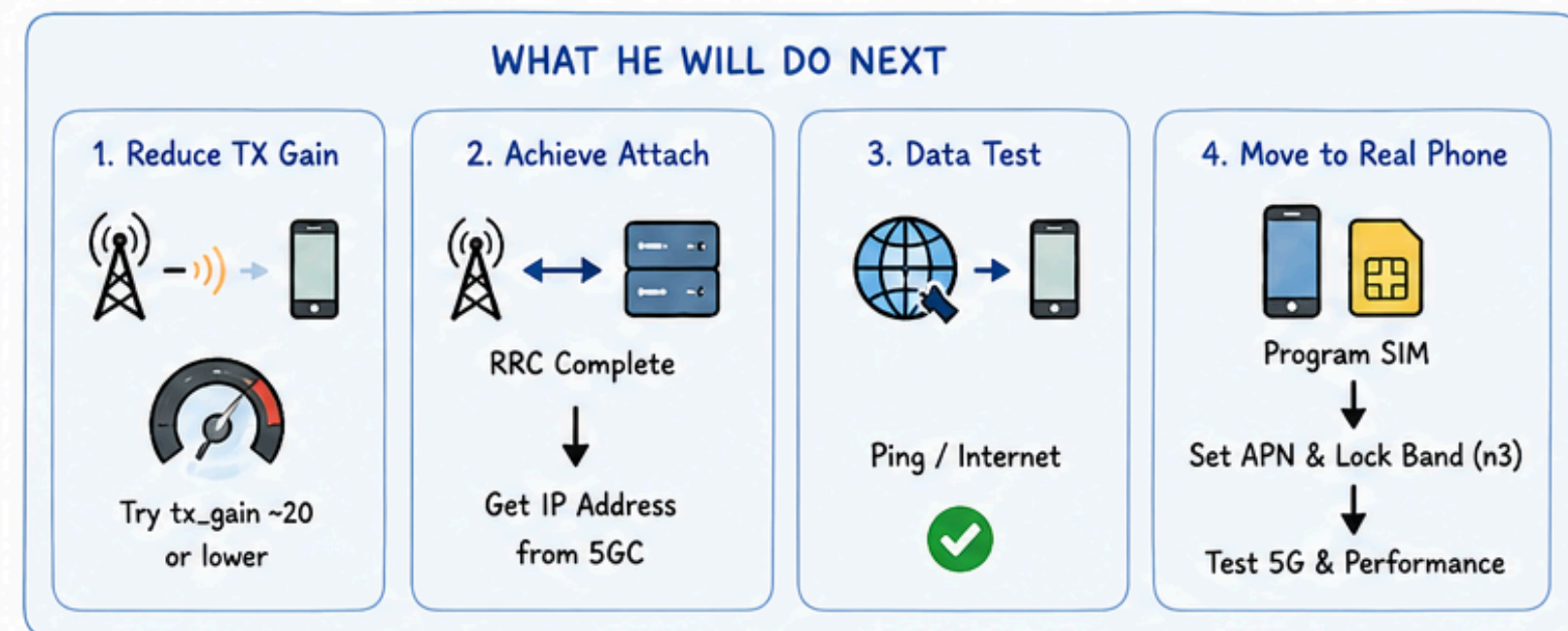
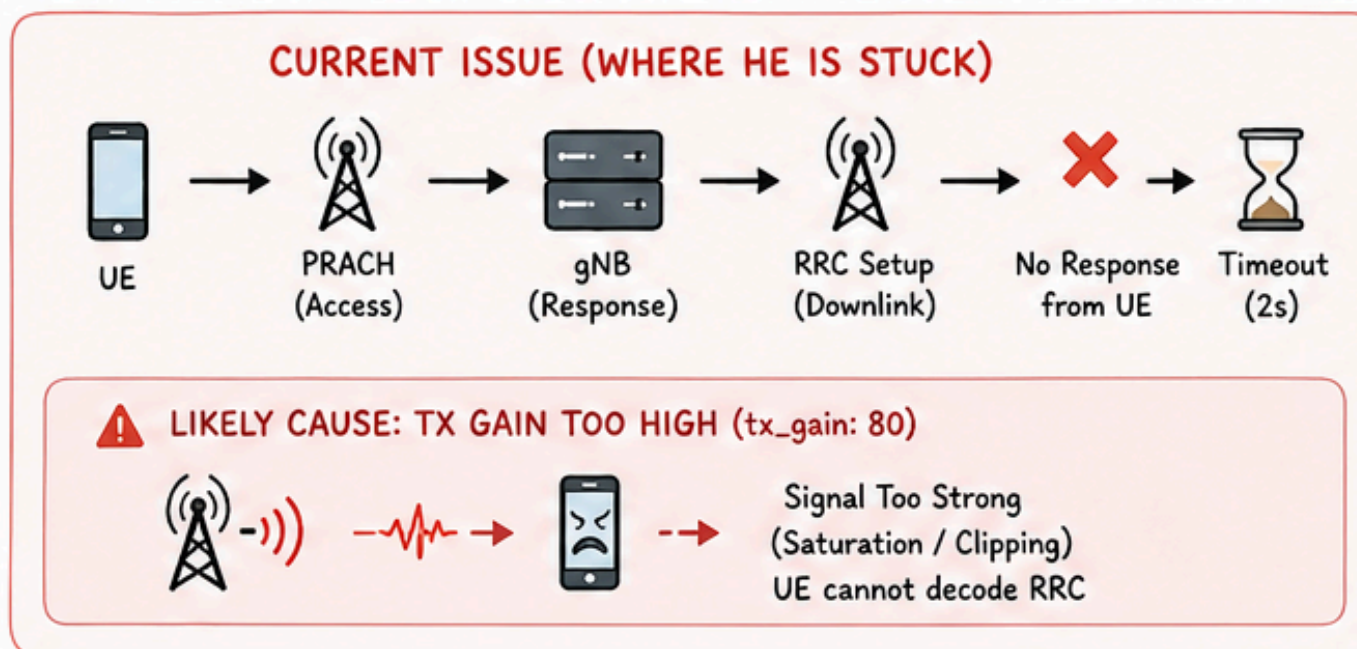
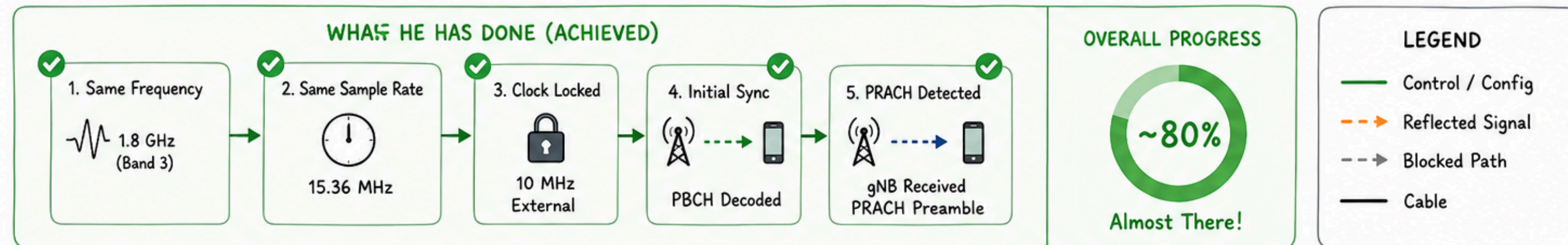
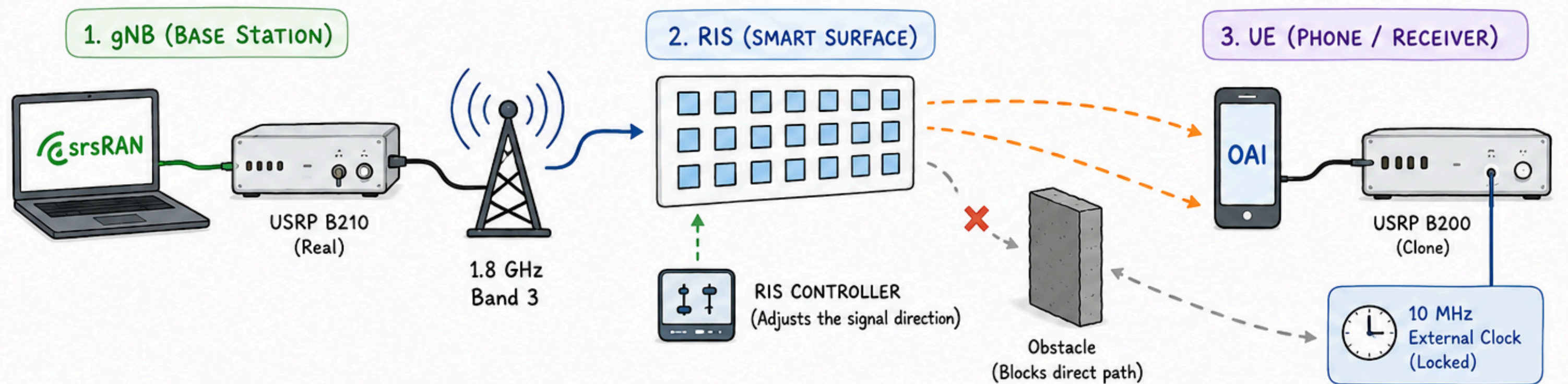
Simulation Duration: 1 h 55 min

Mesh Resolution: 13.24 M Cells

Steady-State Criterion: -40 dB

## SECTION 2

# WORK FLOW



**THANK YOU**